Angle, θ

The angle between two planes means the angle between the normals to the planes

then $$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

where a_1, b_1, c_1 and a_2, b_2, c_2 are direction ratios.

i) If $\theta = 90^\circ$ i.e. perpendicular then

$$\cos 90^\circ = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$\Rightarrow 0 = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$\Rightarrow a_1a_2 + b_1b_2 + c_1c_2 = 0$$

ii) If $\theta = 0^\circ$ i.e. parallel then

$$\cos 0 = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$\Rightarrow \sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2} = a_1a_2 + b_1b_2 + c_1c_2$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Find the angle between the lines

$3x+2y+z-5=0,\ x+y-2z-3=0$

$2x-y-z=0,\ 7x+10y-8z=15$.

Solution:- Given lines

$3x+2y+z-5=0,\ x+y-2z-3=0$ --- (1)

$2x-y-z=0,\ 7x+10y-8z=15$ --- (2)

If l_1, m_1, n_1 be the direction cosines of the line given by (1) then it is perpendicular to normals of both the planes represented by (1). We have

$3l_1+2m_1+n_1=0,\ l_1+m_1-2n_1=0$

Solving these, we get

$$\frac{l_1}{-5}=\frac{m_1}{7}=\frac{n_1}{1}$$

$\therefore$ Direction ratios are $-5, 7, 1 \longrightarrow (a_1, b_1, c_1)$

Again l_2, m_2, n_2 be the direction cosines of then line given by (2) then similarly we get

$2l_2-m_2-n_2=0,\ 7l_2+10m_2-8n_2=0$

Solving these, we get

$$\frac{l_2}{18}=\frac{m_2}{9}=\frac{n_2}{27} \quad \text{or,} \quad \frac{l_2}{2}=\frac{m_2}{1}=\frac{n_2}{3}$$

$\therefore$ Direction ratios are 2, 1, 3. $\longrightarrow$ (a_2, b_2, c_2)

Now

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = -10 + 7 + 3 = 0$$

Hence

$$\cos\theta = 0$$

$$\Rightarrow \theta = 90^\circ$$

i.e. the lines are perpendicular.

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